HIEU N. NGUYEN

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RESEARCH INTERESTS

I'm interested in (i) understanding the fundamental strengths and limitations of learning algorithms through carefully designed, controlled experiments, (ii) developing intelligent systems with capabilities for reasoning, planning, and creativity, and (iii) leveraging these systems to accelerate scientific discovery and tackle real-world challenges at scale.
Research topics: LLM reasoning, Post-training, AI4Science, Interpretability.

EDUCATION

Pennsylvania State University, University Park — *Aug. 2025 - Current*
Ph.D in Computer Science — State College, PA
Advisor: Prof. Rui Zhang
Awarded *University Graduate Fellowship*

Vietnam National University, Hanoi — *Aug. 2022*
B.S Honors Program in Computer Science — Hanoi, Vietnam
Thesis: An application of Neural Tangent Kernel on DEQ Models
Thesis Advisors: Dr. Hoang Thanh-Tung and Dr. Ta Viet Cuong

RESEARCH EXPERIENCE

Ph.D Student, The Pennsylvania State University — *Aug. 2025 – Current*

- Developed a technique to identify critical weights that influence model reasoning behavior.
- Study the impact of training data characteristics on reasoning coverage.

Research Assistant, VinUniversity – CECS — *June 2023 – Aug. 2025*

- Developed LLM-SRBench, a benchmark designed to assess LLMs for scientific equation discovery.
- Mitigated over-optimization in LLM alignment algorithms.
- Enhanced the efficiency and security of federated recommendation systems.

AI Resident, FPT Software AI Center — *May 2021 – June 2023*

- Worked on Out-of-Distribution (OOD) detection, clean-label backdoor attacks, and labeling error detection.

SELECTED PREPRINTS & PUBLICATIONS

*: equal contribution. For my latest research papers, see my Google Scholar profile.

Why Do Reasoning Models Lose Coverage? The Role of Data and Forks in the Road
Hieu N. Nguyen*, Parshin Shojaee*, Phuc Minh Nguyen, Nan Zhang, Chandan K Reddy, Khoa D Doan, and Rui Zhang
arXiv 2026

When Reasoning Meets Compression: Understanding the Effects of LLMs Compression on Large Reasoning Models
Nan Zhang, Eugene Kwek, Yusen Zhang, **Hieu N. Nguyen**, Prasenjit Mitra, and Rui Zhang
ICLR 2026

Wicked Oddities: Selectively Poisoning for Effective Clean-Label Backdoor Attacks
Nguyen Hung-Quang, **Hieu N. Nguyen**, The-Anh Ta, Thanh Nguyen-Tang, Kok-Seng Wong, Hoang Thanh-Tung, and Khoa D Doan
ICLR 2025

Mitigating Reward Over-optimization in Direct Alignment Algorithms with Importance Sampling
Nguyen Minh Phuc, **Hieu N. Nguyen**, Duy Minh Ho Nguyen, Anji Liu, An Mai, Binh T. Nguyen, Daniel Sonntag, and Khoa D Doan
NeurIPS 2025

LLM-SRBench: A New Benchmark for Scientific Equation Discovery with Large Language Models
Parshin Shojaee*, **Hieu N. Nguyen***, Kazem Meidani, Amir Barati Farimani, Khoa D Doan, and Chandan K. Reddy
ICML (Oral presentation - Top 1%) 2025

Towards Efficient Communication and Secure Federated Recommendation System via Low-rank Training
Hieu N. Nguyen, Tuan-Anh Nguyen, Tuan Minh Nguyen, Vu Tien Hoang, Dung D. Le, and Kok-Seng Wong
WWW (Oral presentation) 2024

Class-based Influence Functions for Error Detection
Nguyen-Duc Thang, Hoang Thanh-Tung, Quan Hung Tran, Dang Huu-Tien, **Hieu N. Nguyen**, Anh T. V. Dau, and Nghi D. Q. Bui
Association for Computational Linguistics (ACL) 2023

UNDER REVIEW

*: equal contribution.

Agent-Based Auditing of Mathematical Proofs in Research Papers
Hieu N. Nguyen and Rui Zhang
Submitted to ICML Workshop AI4Research 2026

On the Primacy Bias in RLVR Training
Nguyen Minh Phuc, **Hieu N. Nguyen**, Parshin Shojaee, Chinh Duc La, Duy Minh Ho Nguyen, Rui Zhang, Khoa D. Doan, Ting Hua, and Nitesh V. Chawla
Submitted to NeurIPS 2026

ACADEMIC SERVICES

Reviewer - SPIGM @ ICML 2026, ICLR 2026, ICCV 2025, NeurIPS 2025, ICLR 2025, IEEE Transactions on Image Processing 2025, NeurIPS 2024, AAAI 2024, ICML 2024.

TEACHING

Intro to AI, Teaching Assistant, Spring 2025 — VinUniversity-CECS
Data Mining, Teaching Assistant, Spring 2024, Fall 2024 — VinUniversity-CECS

SKILLS

Programming
Python, C/C++, CUDA.

ML frameworks
PyTorch, Jax, DeepSpeed, FSDP, verl, Unsloth.

DevOps
Linux, Shell (Bash/Zsh), Git, Docker, AWS, Claude Code.